

Simple MVC with Tension PTX



Section I: Scenario Demographics

Scenario Title:	Simple MVC with Tension PTX		
Date of Development:	29/05/2015 (DD/MM/YYYY)		
Target Learning Group:	☒ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to trauma patients that require multiple concurrent management steps.
CRM Objectives:	Effectively lead a trauma team
Medical Objectives:	1) Perform safe primary survey and a proper secondary survey. 2) Recognize and decompress a tension pneumothorax. 3) Appropriately manage an open fracture in a trauma patient.

Case Summary: Brief Summary of Case Progression and Major Events

A 44 year-old male arrives by EMS to a tertiary care ED where the trauma team has been activated. He was the driver in a single-vehicle MVC. He presents screaming and moaning with a GCS of 13. He has an obvious open fracture of his right forearm. He also has decreased air entry to the right side of his chest. The team will need to recognize the tension pneumothorax as part of their primary survey. They will then need to irrigate and splint the right arm after they have completed their secondary survey. As the secondary survey is being completed, the patient will become hypotensive again. This time, the team will find free fluid in the RUQ.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates			
Brief Description of Role			
None.			
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☒ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☒ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: 14G angiocatheter	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Open fracture of forearm.			
E. Approximate Timing			
Set-Up:	6 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

EMS arrives with a 44-year-old male to your tertiary care ED. The trauma team has been activated. He was the driver in a single vehicle MVC at highway speed. There was extensive damage to the car. He is currently screaming and moaning.

B. Patient Profile and History

Patient Name: Joel Brown	Age: 44	Weight: 70kg
Gender: ☒ M ☐ F	Code Status: Full	
Chief Complaint: MVC		

History of Presenting Illness: **Single vehicle MVC at highway speed.**

Past Medical History:	EtOH abuse	Medications:	None.

Allergies: None.

Social History: Unknown.

Family History: Unknown.

Review of Systems:	CNS:	Not cooperative.		
	HEENT:	Nil.		
	CVS:	Some CP if asked.		
	RESP:	Feels SOB if asked.		
	GI:	Nil.		
	GU:	Nil.		
	MSK:	Screams when move R arm.	INT:	Nil.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 130/min	BP: 75/35	RR: 26/min
O ₂ SAT: 94%		
Rhythm: Sinus tach	T: 36.2°C	Glucose: 6.4 mmol/L
GCS: 13 (E4 V4 M5)		
General Status: Appears in pain. Agitated.		
CNS:	Not obeying commands. Will answers simple yes/no questions. Not oriented.	
HEENT:	PERLA, 4mm. No signs HI.	
CVS:	Normal S1/S2. No murmurs.	
RESP:	No AE to R side.	
ABDO:	Soft, NT.	
GU:	Nil.	
MSK:	Pelvis stable. Open # to R forearm.	SKIN: Nil else.

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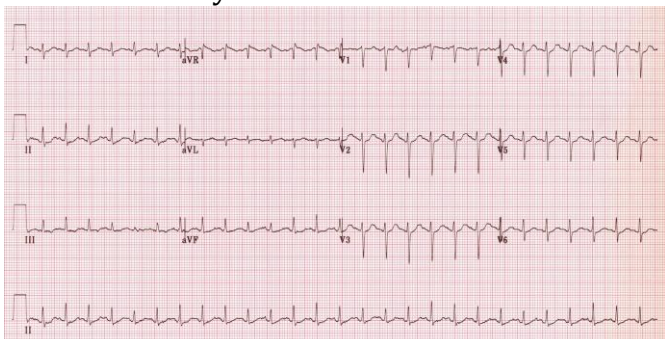
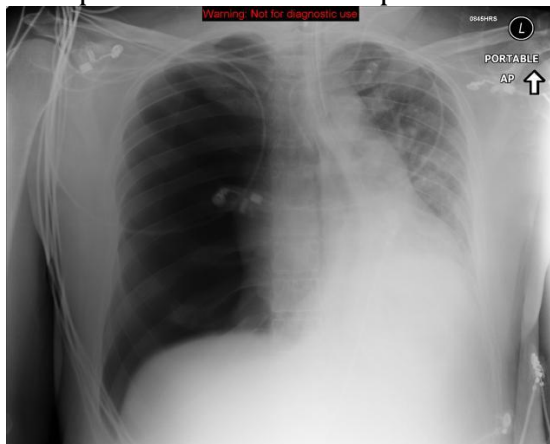
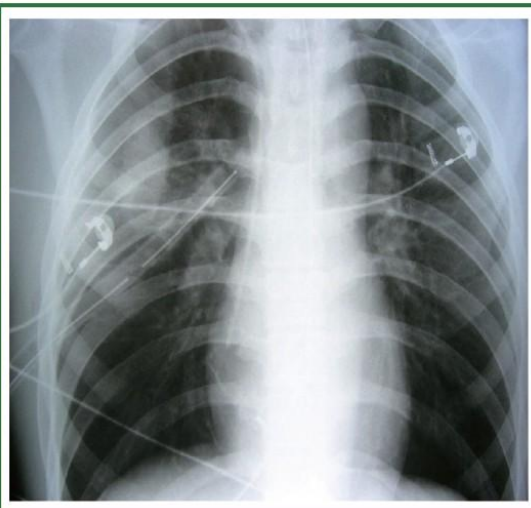
Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 130/min BP: 75/35 RR: 26/min O ₂ SAT: 94% T: 36.2°C	Patient is screaming and moaning, not obeying commands. Answers simple questions.	<u>Learner Actions</u> - ☐ ABC, monitors, iv access x2 - ☐ Bolus IVF, call for blood - ☐ Blood work (inc. G&S) - ☐ Needle decompress chest - ☐ ± Chest tube insertion - ☐ Call for CXR/PXR - ☐ Analgesia (fentanyl or ketamine iv) - ☐ Check cap sugar (6.4) - ☐ Administer TXA 1g iv	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - No needle decompression by 3 minutes → BP 60/30, O ₂ SAT 90%, HR 140 <u>Triggers</u> <i>For progression to next state</i> - Needle decompression → 2. Stabilization
2. Stabilization HR → 110 BP → 95/60	Patient still moans, responds. But less wild after analgesia.	<u>Learner Actions</u> - ☐ Chest tube insertion - ☐ CXR/PXR - ☐ Complete px exam - ☐ FAST to r/o FF (+ RUQ) - ☐ Td 0.5 ml IM - ☐ Ancef 2g iv ± gentamycin iv - ☐ Irrigate and splint open #	<u>Modifiers</u> - No lidocaine for chest tube insertion → patient screams ++ <u>Triggers</u> - Chest tube inserted or FAST complete → 3. Repeat hypotension - 8 minutes → 3. Repeat hypotension
3. Repeat hypotension HR → 125 BP → 75/35	As above.	<u>Learner Actions</u> - ☐ Re-check air entry - ☐ FAST to r/o FF (+ RUQ) - ☐ Pericardial U/S to r/o tamponade - ☐ Assess chest tube drainage, consider repeat CXR or U/S - ☐ Administer 2u PRBC - ☐ Administer TXA 1g iv (if not yet done) - ☐ Call general surgery	<u>Modifiers</u> - 1 unit PRBC → BP 80/40 - No FAST by 10 min → BP 70/30 - No FAST by 11 min → BP 65/30 <u>Triggers</u> - 2 units PRBC and surgery called → 4. Resolution - 12 minutes → 4. Resolution
4. Resolution HR → 115 BP → 85/45	As above.	<u>Learner Actions</u> - ☐ Consider intubation for OR - ☐ Call consultants	END CASE → GEN SX ARRIVES

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results	
None given during scenario.	
Images (ECGs, CXRs, etc.)	
ECG: sinus tachycardia  ECG source: https://lifeinthefastlane.com/ecg-library/sinus-tachycardia/	CXR: prior to needle decompression  CXR source: https://radiopaedia.org/cases/tension-pneumothorax-9
Pelvic xray: normal  PXR source: http://radiopaedia.org/articles/pelvis-1	CXR: post chest tube placement  CXR source: http://jtd.amegroups.com/article/view/663/html
Ultrasound Video Files (if applicable)	
U/S RUQ showing free fluid	
U/S pericardium showing no effusion	

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To expose learners to trauma patients that require multiple concurrent management steps.
CRM Objectives:	Effectively lead a trauma team
Medical Objectives:	1) Perform safe and thorough primary survey. 2) Recognize and decompress a tension pneumothorax. 3) Appropriately manage an open fracture in a trauma patient.
Sample Questions for Debriefing	
1) When did you recognize the tension pneumothorax? 2) What is your approach to fluid resuscitation in trauma? 3) What approach did the team leader use to organizing the initial management of this trauma? Does ABC (with stops for interventions) work or should we be completing a primary survey and then delegating tasks from there? 4) How did the mood of the room feel? Did the team feel and recognize when the patient was worsening?	
Key Moments	
1) Recognition of PTX	
2) Completing secondary survey	
3) Repeating primary survey when patient's BP worsened	